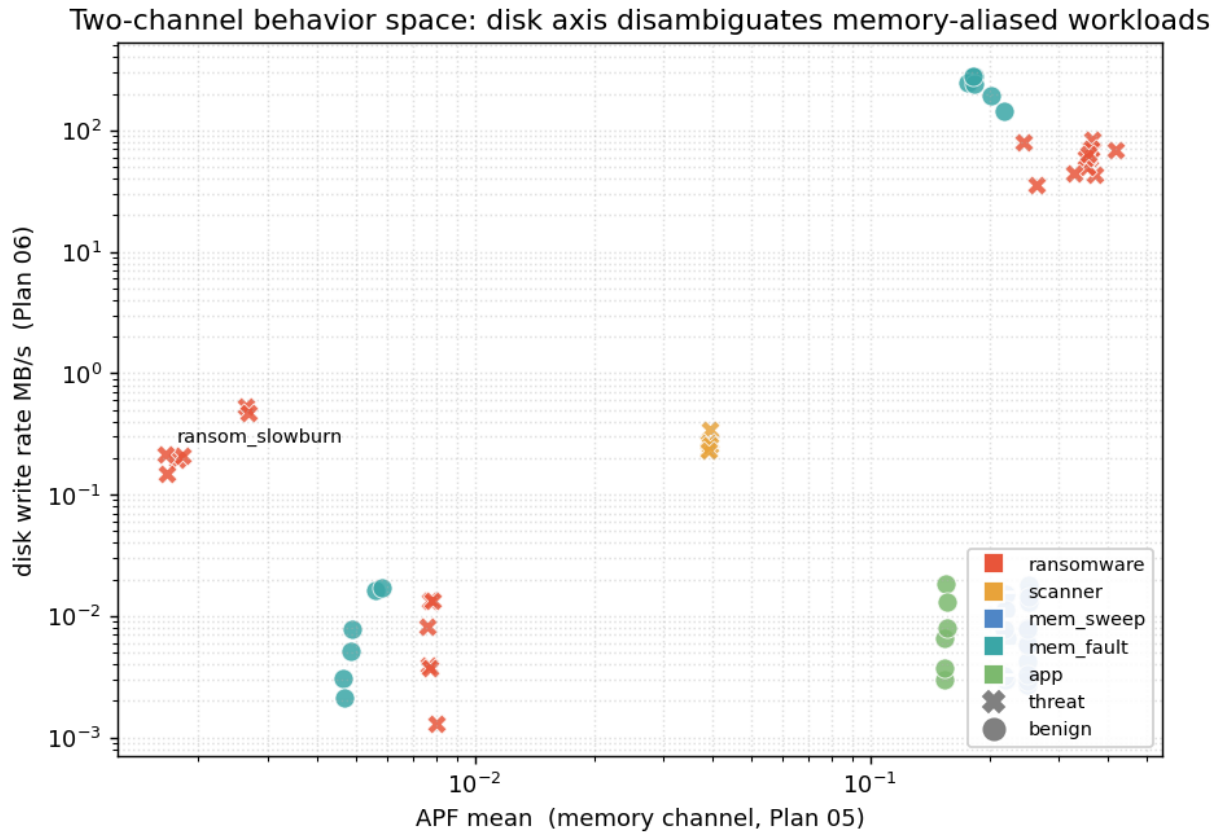


Plan 05 → Plan 06 — Comprehensive Data

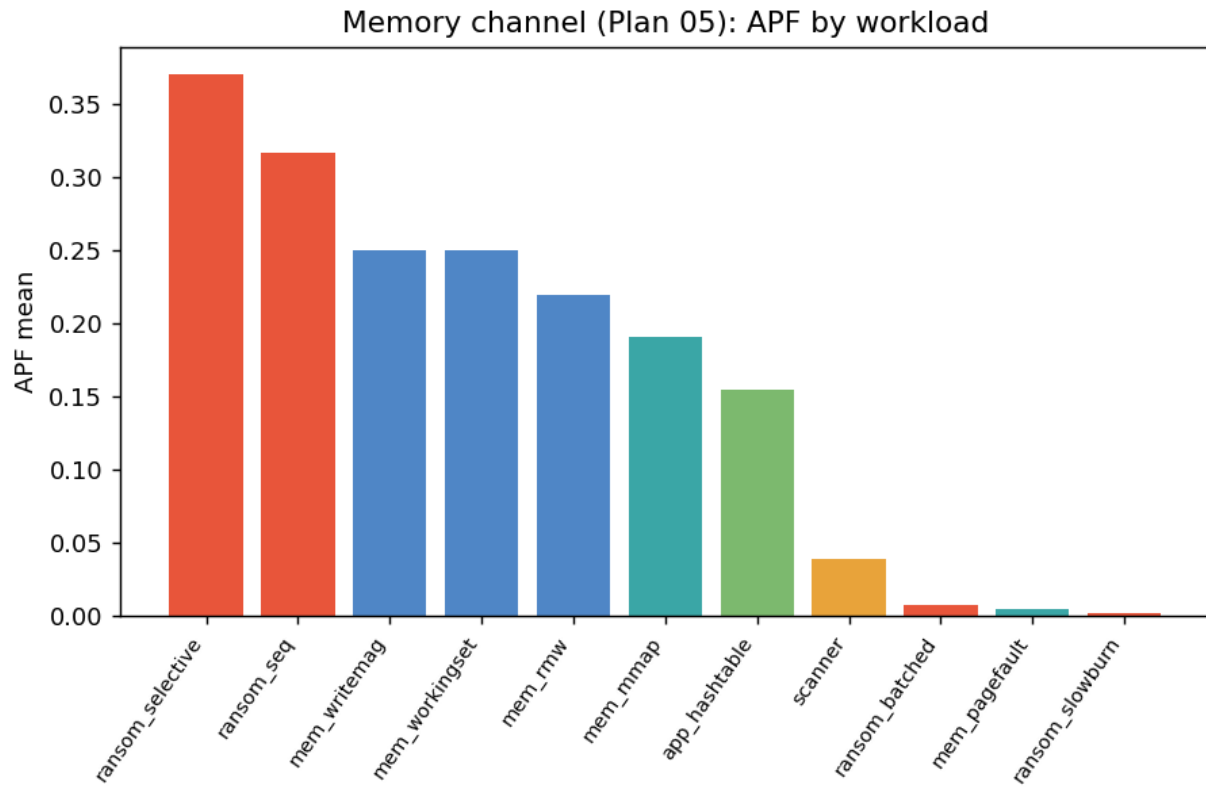
All 66 cells, 11 workloads, both channels. Memory (APF) = Plan 05; disk = Plan 06.

1. Two-channel behavior space

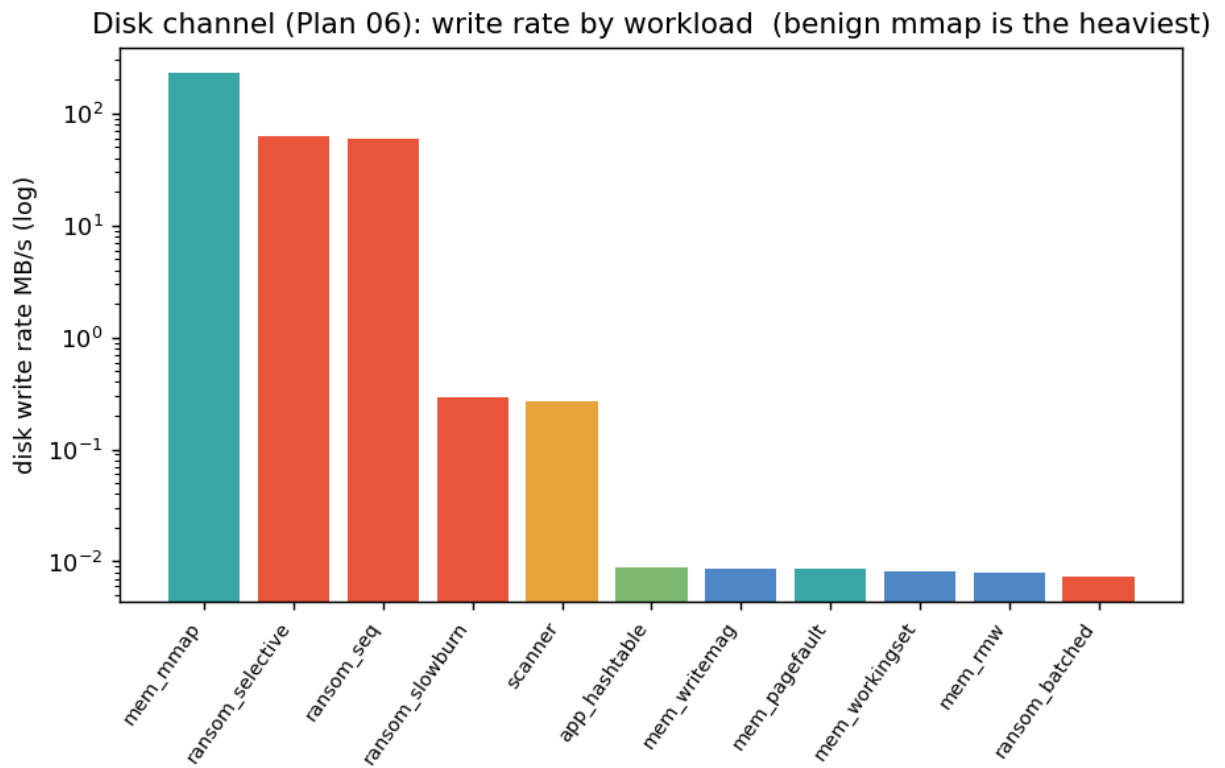


Each point = one cell. X = APF mean (memory), Y = disk write rate (Plan 06). Memory-aliased workloads (slowburn vs pagefault) separate on the disk axis; the heaviest writer is benign (mmap).

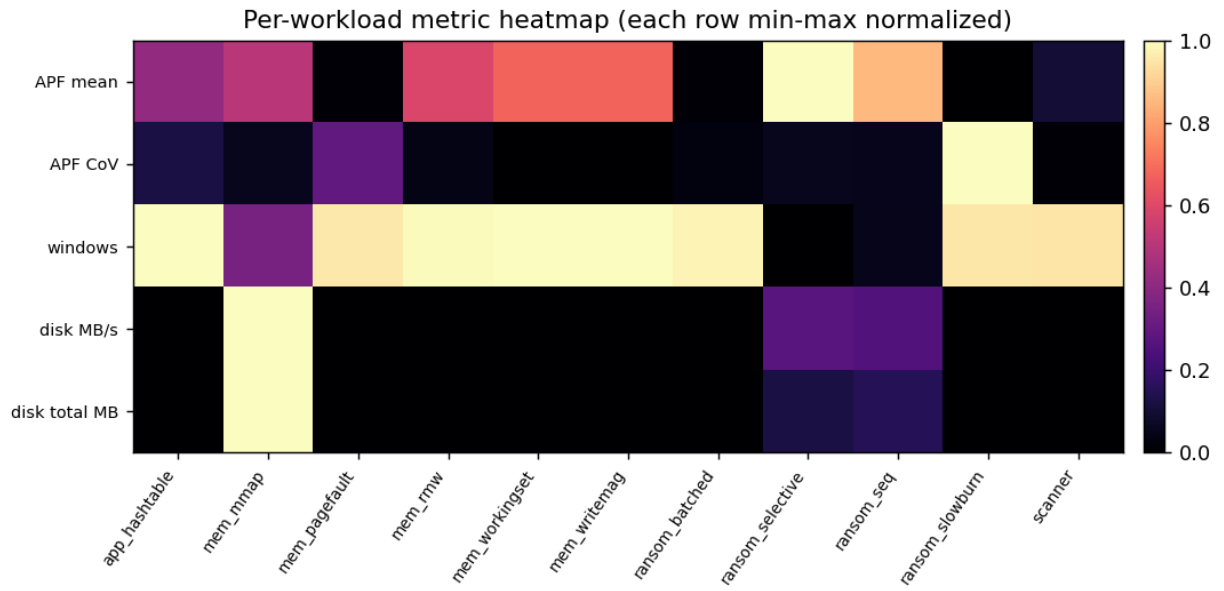
2. Per-workload, both channels



Memory: APF mean per workload.



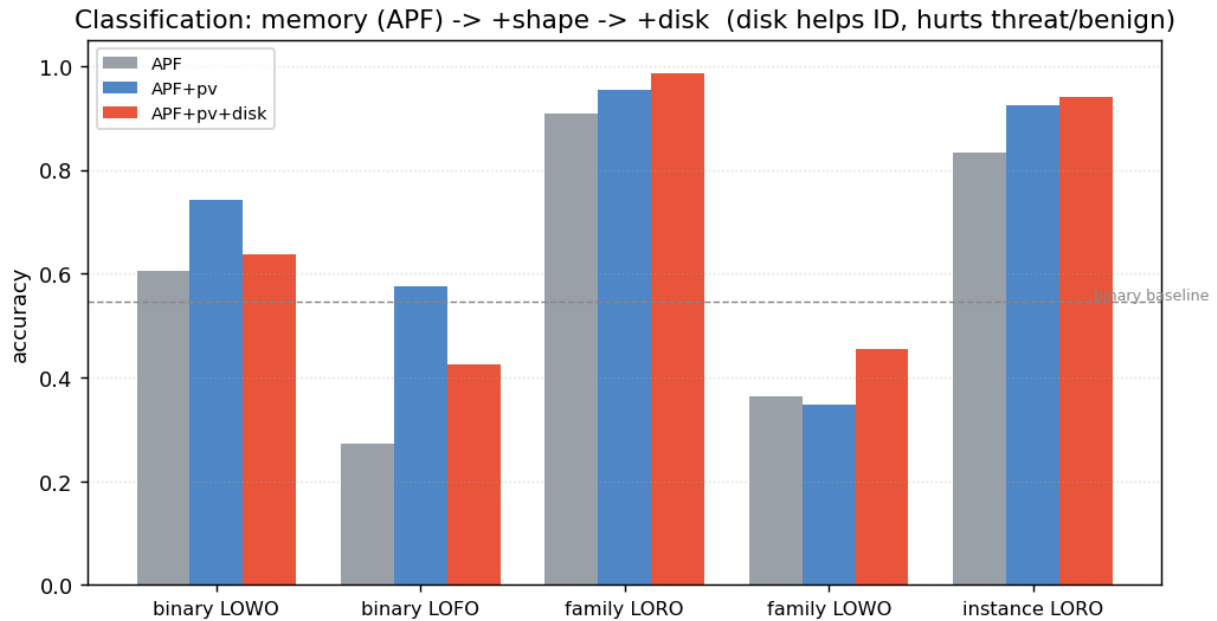
Disk: write rate per workload (log).



Per-workload metric heatmap (rows min-max normalized).

family	workload	kind	APF mean	APF CoV	disk MB/s	disk MB	win
app	app_hashtable_intensive	benign	0.1550	0.75	0.009	9	138
mem_fault	mem_mmap_traversal	benign	0.1907	0.36	231.636	209083	56
mem_fault	mem_pagefault_density	benign	0.0051	1.80	0.009	9	132
mem_sweep	mem_rmw_intensity	benign	0.2193	0.25	0.008	9	137
mem_sweep	mem_workingset_sweep	benign	0.2507	0.01	0.008	9	137
mem_sweep	mem_writemag_sweep	benign	0.2507	0.01	0.009	8	138
ransomware	ransom_batched	threat	0.0077	0.18	0.007	9	135
ransomware	ransom_selective	threat	0.3710	0.35	62.519	26126	12
ransomware	ransom_seq	threat	0.3172	0.32	59.070	31500	18
ransomware	ransom_slowburn	threat	0.0020	6.05	0.293	369	131
scanner	scanner_metadata	threat	0.0392	0.08	0.269	363	131

3. Classification: what each channel buys



Accuracy by feature set across tasks. Disk helps identification, hurts threat/benign detection.

task	APF only	+peak/var	+disk
binary LOWO	0.606	0.742	0.636
binary LOFO	0.273	0.576	0.424
family LORO	0.909	0.955	0.985
family LOWO	0.364	0.348	0.455
instance LORO	0.833	0.924	0.939

Disk I/O helps characterization (family LORO 0.955→0.985, family LOWO 0.348→0.455, instance 0.924→0.939) and hurts detection (binary LOWO 0.742→0.636) — a characterization channel, not a detector.

4. Every cell (66)

fam	workload	kind	dur	rep	APF	CoV	win	MB/s	MB
app	app_hashtable_intensive	benign	120	1	0.1552	0.75	48	0.018	10
app	app_hashtable_intensive	benign	120	2	0.1570	0.74	48	0.013	7
app	app_hashtable_intensive	benign	300	1	0.1543	0.76	121	0.006	9
app	app_hashtable_intensive	benign	300	2	0.1560	0.75	121	0.008	11
app	app_hashtable_intensive	benign	600	1	0.1540	0.76	244	0.003	8
app	app_hashtable_intensive	benign	600	2	0.1535	0.76	244	0.004	9
mem_fault	mem_mmap_traversal	benign	120	1	0.1828	0.38	36	277.986	142101
mem_fault	mem_mmap_traversal	benign	120	2	0.1760	0.40	10	248.990	38693
mem_fault	mem_mmap_traversal	benign	300	1	0.1838	0.40	89	243.891	333324
mem_fault	mem_mmap_traversal	benign	300	2	0.1817	0.40	92	283.370	382741
mem_fault	mem_mmap_traversal	benign	600	1	0.2022	0.32	89	192.433	291447
mem_fault	mem_mmap_traversal	benign	600	2	0.2176	0.25	21	143.144	66193
mem_fault	mem_pagefault_density	benign	120	1	0.0056	2.19	47	0.016	8
mem_fault	mem_pagefault_density	benign	120	2	0.0058	2.59	47	0.017	9
mem_fault	mem_pagefault_density	benign	300	1	0.0049	1.61	121	0.005	7
mem_fault	mem_pagefault_density	benign	300	2	0.0049	1.75	121	0.008	10
mem_fault	mem_pagefault_density	benign	600	1	0.0046	1.34	243	0.003	8
mem_fault	mem_pagefault_density	benign	600	2	0.0047	1.32	212	0.002	13
mem_sweep	mem_rmw_intensity	benign	120	1	0.2190	0.27	47	0.015	9
mem_sweep	mem_rmw_intensity	benign	120	2	0.2190	0.24	47	0.011	9
mem_sweep	mem_rmw_intensity	benign	300	1	0.2211	0.24	121	0.007	9
mem_sweep	mem_rmw_intensity	benign	300	2	0.2183	0.25	121	0.008	10
mem_sweep	mem_rmw_intensity	benign	600	1	0.2203	0.24	243	0.003	8
mem_sweep	mem_rmw_intensity	benign	600	2	0.2180	0.26	243	0.003	8
mem_sweep	mem_workingset_sweep	benign	120	1	0.2510	0.01	48	0.015	8
mem_sweep	mem_workingset_sweep	benign	120	2	0.2509	0.01	47	0.013	8
mem_sweep	mem_workingset_sweep	benign	300	1	0.2506	0.01	121	0.008	11
mem_sweep	mem_workingset_sweep	benign	300	2	0.2507	0.01	121	0.006	10
mem_sweep	mem_workingset_sweep	benign	600	1	0.2506	0.00	244	0.004	11
mem_sweep	mem_workingset_sweep	benign	600	2	0.2506	0.00	243	0.003	8
mem_sweep	mem_writemag_sweep	benign	120	1	0.2509	0.01	48	0.018	9
mem_sweep	mem_writemag_sweep	benign	120	2	0.2509	0.01	48	0.015	7
mem_sweep	mem_writemag_sweep	benign	300	1	0.2507	0.01	122	0.006	8
mem_sweep	mem_writemag_sweep	benign	300	2	0.2507	0.01	121	0.008	10
mem_sweep	mem_writemag_sweep	benign	600	1	0.2506	0.00	244	0.003	8
mem_sweep	mem_writemag_sweep	benign	600	2	0.2507	0.01	243	0.003	9
ransomware	ransom_batched	threat	120	1	0.0077	0.23	45	0.013	7
ransomware	ransom_batched	threat	120	2	0.0078	0.25	47	0.013	7
ransomware	ransom_batched	threat	300	1	0.0076	0.16	121	0.008	11
ransomware	ransom_batched	threat	300	2	0.0076	0.17	115	0.004	9

fam	workload	kind	dur	rep	APF	CoV	win	MB/s	MB
ransomware	ransom_batched	threat	600	1	0.0077	0.14	242	0.004	10
ransomware	ransom_batched	threat	600	2	0.0080	0.13	240	0.001	8
ransomware	ransom_selective	threat	120	1	0.3620	0.41	5	69.990	13373
ransomware	ransom_selective	threat	120	2	0.3643	0.37	7	83.080	15213
ransomware	ransom_selective	threat	300	1	0.3580	0.35	14	59.624	29766
ransomware	ransom_selective	threat	300	2	0.4167	0.25	11	68.899	24021
ransomware	ransom_selective	threat	600	1	0.3699	0.32	23	43.678	46233
ransomware	ransom_selective	threat	600	2	0.3548	0.42	15	49.841	28152
ransomware	ransom_seq	threat	120	1	0.2436	0.29	12	79.629	17276
ransomware	ransom_seq	threat	120	2	0.3627	0.29	9	71.480	18292
ransomware	ransom_seq	threat	300	1	0.3503	0.30	11	59.579	20552
ransomware	ransom_seq	threat	300	2	0.3563	0.28	15	63.480	29877
ransomware	ransom_seq	threat	600	1	0.3272	0.36	26	44.903	43017
ransomware	ransom_seq	threat	600	2	0.2632	0.41	38	35.348	59985
ransomware	ransom_slowburn	threat	120	1	0.0026	6.19	45	0.528	260
ransomware	ransom_slowburn	threat	120	2	0.0027	6.08	46	0.467	260
ransomware	ransom_slowburn	threat	300	1	0.0018	5.95	115	0.194	328
ransomware	ransom_slowburn	threat	300	2	0.0018	5.79	115	0.207	326
ransomware	ransom_slowburn	threat	600	1	0.0017	6.19	234	0.215	601
ransomware	ransom_slowburn	threat	600	2	0.0017	6.12	233	0.149	439
scanner	scanner_metadata	threat	120	1	0.0394	0.08	48	0.277	144
scanner	scanner_metadata	threat	120	2	0.0392	0.10	24	0.340	119
scanner	scanner_metadata	threat	300	1	0.0390	0.07	122	0.238	308
scanner	scanner_metadata	threat	300	2	0.0390	0.08	112	0.269	343
scanner	scanner_metadata	threat	600	1	0.0394	0.07	235	0.257	658
scanner	scanner_metadata	threat	600	2	0.0391	0.06	244	0.234	605